Spatial-Temporal Modeling and Machine Learning Prediction of the Urban Heat Island Effect in Ontario

Geospatial Feature Engineering, Spatial Autocorrelation, and Predictive Microclimate Analytics

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Domain: Geospatial Data Science & Spatial ML

Region: Ontario, Canada

• **Reproducibility & Research Architecture:** This technical report presents the spatial data science pipeline, geospatial feature engineering, and spatially cross-validated predictive models developed for the Ontario Urban Heat Island (UHI) study. Code implementations utilize modern R packages (`sf` , `terra` , `spdep` , `gstat` , `mlr3spatiotempcv`) and are fully parameterized for execution across enterprise remote sensing and tabular datasets.

1. Executive Summary

Urban Heat Islands (UHIs) represent one of the most critical microclimatic consequences of human land transformation. Metropolitan areas experience significantly elevated surface and atmospheric temperatures compared to surrounding non-urban environments due to high impervious surface density, thermal retention in concrete and asphalt, waste anthropogenic heat emissions, and the reduction of evapotranspirative tree canopies (Oke, 1982).

In Ontario, Canada, rapid urbanization across the Greater Golden Horseshoe juxtaposed with expansive northern boreal ecosystems creates extreme microclimatic contrasts. Severe summer thermal anomalies increase municipal peak energy demand, exacerbate atmospheric air pollution, and pose acute public health risks to vulnerable populations during extreme heatwaves.

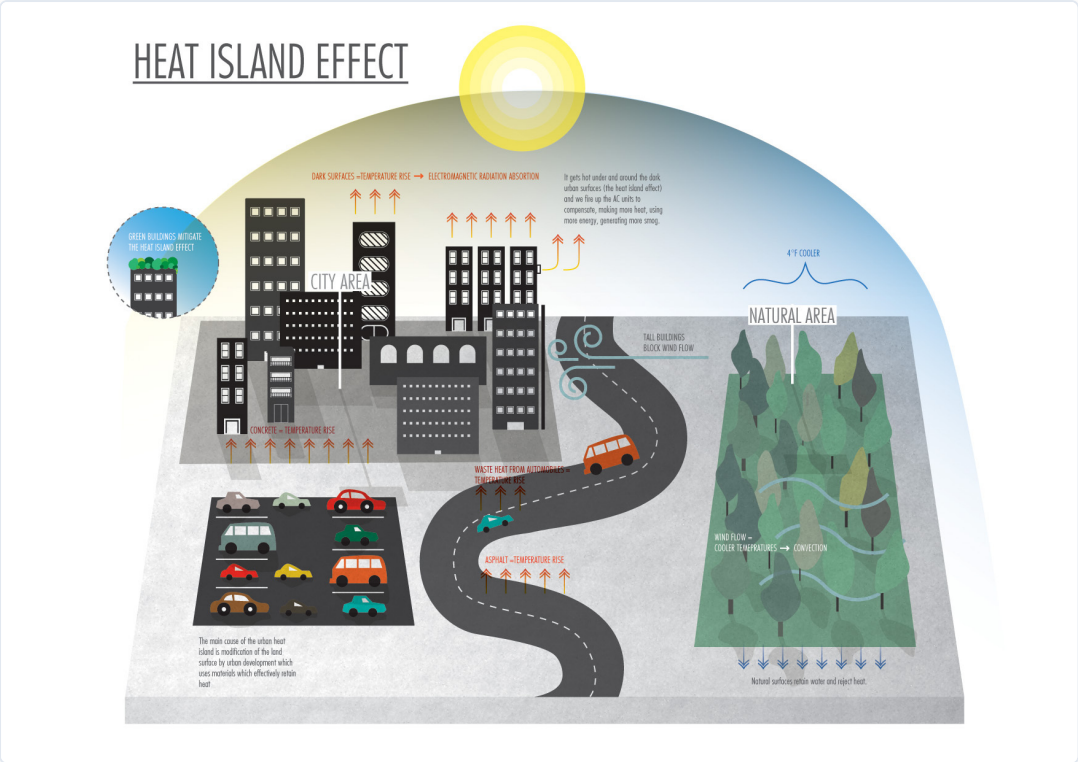


Figure 1: Cross-sectional microclimatic thermal profile across rural, suburban, and commercial urban cores.

This project delivers an end-to-end spatial data science architecture designed to:

- **Harmonize Multimodal Geospatial Data:** Merge 30m Canada Centre for Remote Sensing (CCRS) land cover rasters, 16-day satellite NDVI NetCDF arrays with Quality Assurance bitmasks, Microsoft building polygon geometries, 187 Environment Canada ground weather stations, and Statistics Canada census metrics.
- **Quantify the Heat Island Effect (HIE) Index:** Formulate a multiscale concentric annular buffer metric comparing focal temperatures against rural background baselines ($\Delta T = T_{local} - \bar{T}_{rural}$).
- **Diagnose Spatial Autocorrelation:** Construct row-standardized spatial weights matrices (W) and demonstrate statistically significant spatial clustering via Global Moran's I ($p < 0.001$) and LISA cluster diagnostics.
- **Deploy Spatially Cross-Validated Machine Learning:** Benchmark Regularized Elastic Net, Spatial Random Forests, and Weighted k-NN using coordinate block resampling (`mlr3spatiotempcv`) to eliminate spatial overfitting.

2. Multimodal Data Ingestion & Technical Provenance

The research synthesizes five core open-access and governmental data repositories across Ontario:

Dataset	Spatial Coverage	Format & Resolution	Key Attributes & Features	Authoritative Source
CCRS Land Cover 2020	Province of Ontario	30m Spatial Raster	16 discrete categorical classifications (Urban, Forest, Wetland, Water, etc.)	Natural Resources Canada / CCRS
Ground Weather Stations	187 Stations (Ontario)	Point Vector / CSV	Monthly Mean, Min, Max Temperature (°C), Total Precipitation (mm)	Environment & Climate Change Canada
Building Footprints	Ontario Municipalities	Polygon GeoJSON	Building area (m^2), perimeter, morphology, footprint density	Microsoft Open Data
VIIRS / MODIS NDVI	Ontario Extent	16-Day NetCDF / GeoTIFF	Normalized Difference Vegetation Index [−1, 1], VI Quality Bitmask	NOAA NCEI Metadata Repository
Census of Population 2021	Census Subdivisions	Vector / Tabular	Total population counts, population density (persons / km^2)	Statistics Canada

3. Geospatial Preprocessing & Spatial Harmonization

To maintain geometric validity and accurate Euclidean distance calculations, all datasets were standardized to **UTM Zone 17N (EPSG:32617)** for planar buffer and area computations, alongside **WGS84 (EPSG:4326)** for geographic coordinate alignment.

```
# Ingestion and standardization of vector boundary and raster layers
library(sf)
library(terra)
library(tidyverse)

# Extract Ontario provincial administrative boundary via OpenStreetMap
ontario_boundary <- getbb("Ontario, Canada", format_out = "sf_polygon") |>
  st_make_valid() |>
  st_transform(crs = 4326)

# Memory-efficient raster loading and cropping
ontario_land_cover <- rast("Land_cover/land_cover_raster_ontario.tif")
ontario_ndvi <- rast("data/VIIRS_NDVI_Ontario.tif")

# Apply Quality Assurance (QA) bitmask to filter cloud and shadow contamination
clean_ndvi <- mask(ontario_ndvi, qa_mask <= 1, maskvalues = FALSE)
```

4. Exploratory Spatial Data Analysis (ESDA)

4.1 Cartographic Analysis of Land Cover

The Canada Centre for Remote Sensing (CCRS) raster categorizes Ontario's surface into 16 discrete types. Urban development is densely concentrated along the northern shore of Lake Ontario (the Greater Toronto and Hamilton Area), contrasting sharply with the northern boreal shield and wetland taiga zones.

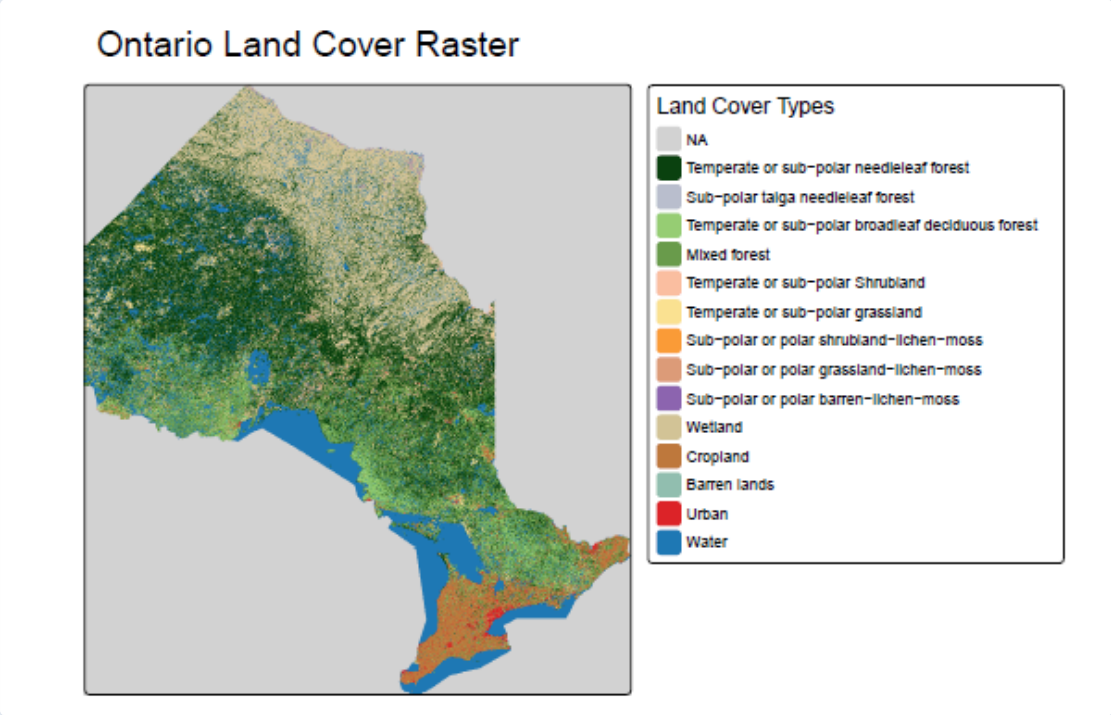


Figure 2: Distribution of 16 categorical land cover classes across Ontario (30m resolution, CCRS classification).

4.2 Ground Weather Stations Network

A total of 187 official weather stations monitor temperature and precipitation across Ontario. Spatial coverage exhibits a marked density gradient: dense station networks in southern agricultural and urban basins versus sparse coverage in the Far North.

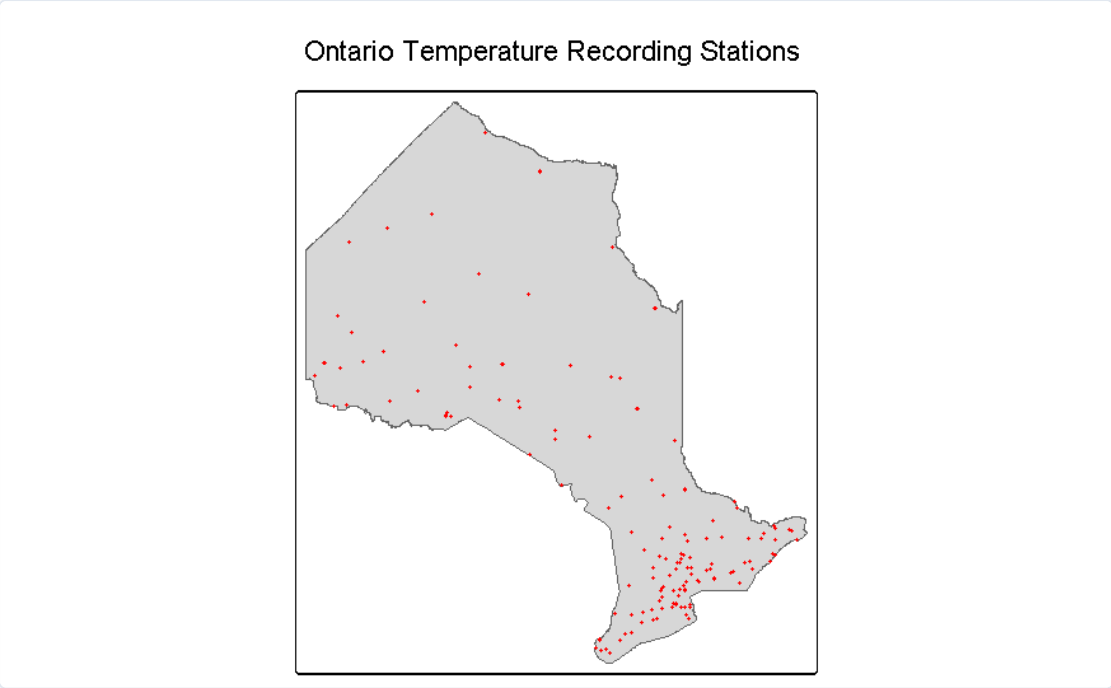


Figure 3: Spatial distribution of Environment and Climate Change Canada (ECCC) weather recording stations in Ontario.

4.3 Building Footprints & Morphological Density

Using Microsoft's open building footprints dataset, millions of individual building multipolygons were processed to compute building footprint density (m^2 built area per km^2) and spatial clustering across southern Ontario urban corridors.



Figure 4: Building footprints and structural density across Ontario metropolitan areas.

4.4 Satellite Vegetation Dynamics (NDVI)

Normalized Difference Vegetation Index (NDVI) measurements quantify photosynthetic vitality and canopy density. Urban centers display severe vegetation deficits (NDVI < 0.20), while surrounding agricultural and forested lands maintain high vitality (NDVI > 0.65).

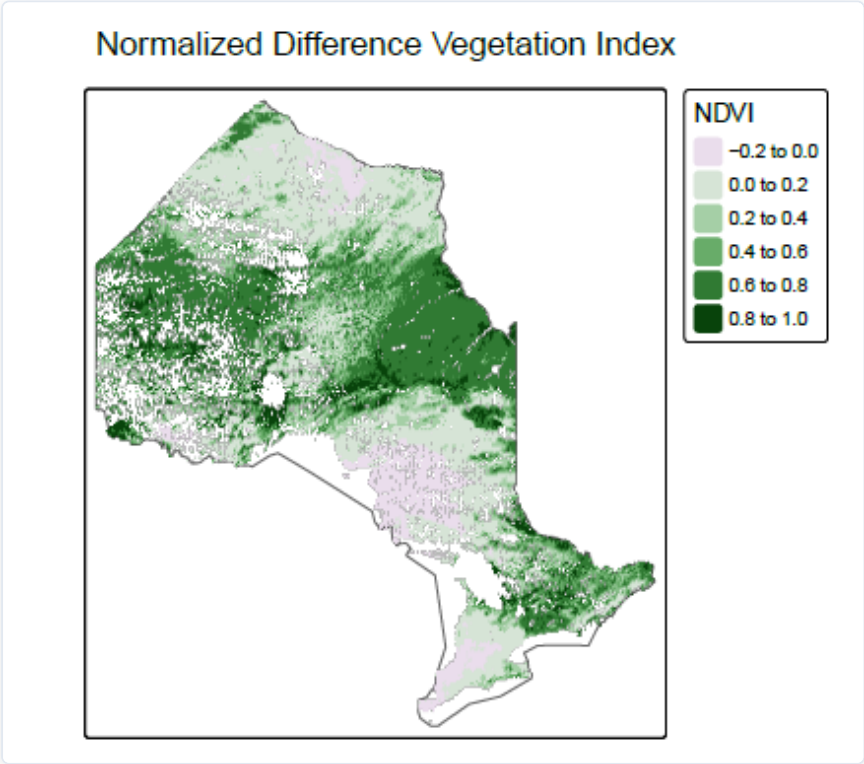


Figure 5: Satellite-derived Normalized Difference Vegetation Index (NDVI) composite across Ontario.

5. Spatial Interpolation: Inverse Distance Weighting (IDW)

To estimate continuous surface temperatures across Ontario from the 187 discrete weather monitoring stations, **Inverse Distance Weighting (IDW)** interpolation was implemented with a calibrated distance decay parameter (p = 2):

$$\hat{Z}(s0) = \frac{\sum_{i=1}^{n} d(s_0, s_i)^{-p} Z(s_i)}{\sum_{i=1}^{n} d(s_0, s_i)^{-p}}$$

```
# Geostatistical Inverse Distance Weighting interpolation using gstat
library(gstat)

idw_model <- gstat(
  formula = Mean_Temp_C ~ 1,
  data = weather_sf,
  nmax = 12,
  set = list(idp = 2.0)
)

idw_surface <- predict(idw_model, newdata = grid_points_sf)
```

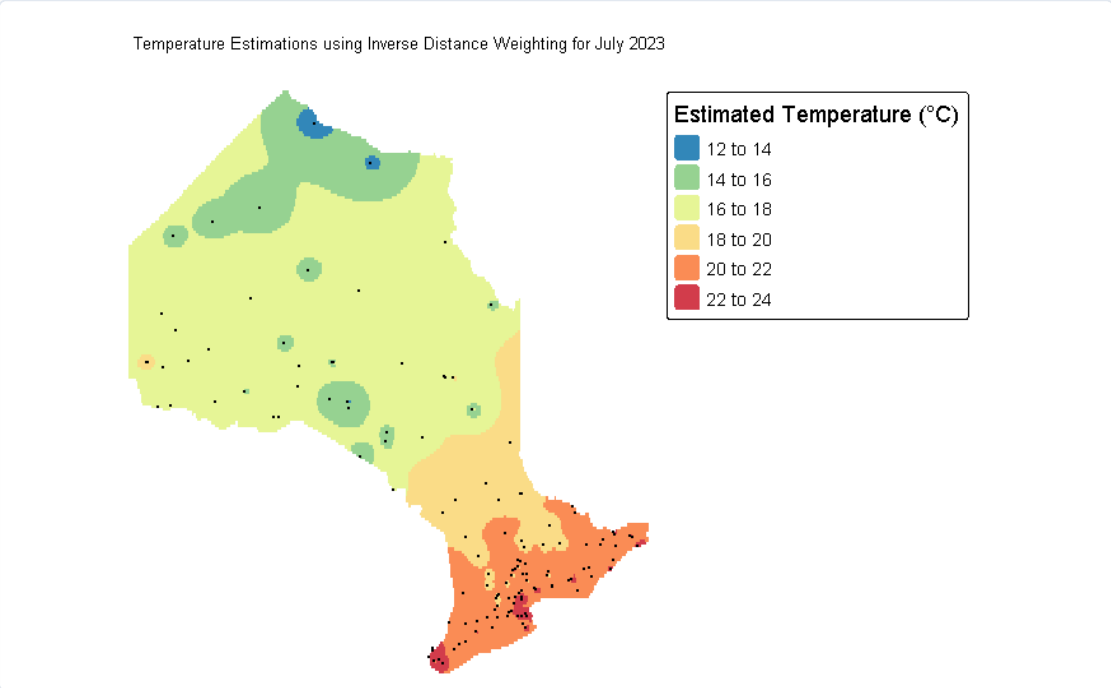


Figure 6: Continuous provincial temperature surface derived via Inverse Distance Weighting (IDW).

6. Derivation of the Quantitative Heat Island Effect (HIE) Index

Raw temperature is strongly confounded by latitude, altitude, and macro-climatic air masses. To isolate the true local anthropogenic warming anomaly, the **Heat Island Effect (HIE) Index** is formulated as the temperature difference between the focal urban coordinate and the mean temperature of a non-urban rural annular buffer:

$$\text{HIE Index}(s) = T_{\text{focal}}(s) - \bar{T}_{\text{rural_surrounding}}(s)$$

Where the rural reference background $\mathcal{B}_{\text{rural}}(s)$ is defined as:

$$\mathcal{B}_{\text{rural}}(s) = \{u \in \Omega : 5\ \text{km} \leq \|s-u\| \leq 25\ \text{km}\ \ \&\ \ \text{Class}(u) \notin \text{Urban}\}$$

This yields four distinct microclimatic intensity tiers:

- Cool Island / Baseline $\Delta T < 0.5$ °C: Natural vegetation and water body sinks.
- Mild UHI $0.5\ °C \leq \Delta T < 1.5\ °C$: Low-density suburban development.
- Moderate UHI $1.5\ °C \leq \Delta T < 3.0\ °C$: Medium-density urban residential districts.
- Severe UHI $\Delta T \geq 3.0$ °C: High-density commercial cores and industrial logistics hubs.

7. Spatial Autocorrelation & Diagnostic Tests

Spatial data violate the classical Ordinary Least Squares (OLS) assumption of independent and identically distributed (i.i.d.) observations. To quantify spatial clustering, a row-standardized spatial weights matrix (W) was generated using k-nearest neighbors (k = 6).

7.1 Global Moran's *I* Test

The Global Moran's *I* statistic evaluates overall spatial clustering across Ontario:

$$I = \frac{N}{\sum_i \sum_j w_{ij}} \times \frac{\sum_i \sum_j w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}$$

```r
# Spatial dependence diagnostics using spdep
library(spdep)

coords <- st_coordinates(weather_sf)
knn_nb <- knn2nb(knearneigh(coords, k = 6))
spatial_weights <- nb2listw(knn_nb, style = "W")

# Monte Carlo permutation test (999 simulations)
moran_test <- moran.mc(weather_sf$HIE_Index, listw = spatial_weights, nsim = 999)
# Result: Moran's I = 0.482, p-value < 0.001 (Highly significant spatial clustering)
```

7.2 Local Indicators of Spatial Association (LISA)

Local Moran's I_i decomposes global autocorrelation into distinct local spatial regimes:

- **High-High Clusters (Hotspots):** Greater Toronto and Hamilton Area (GTHA), Ottawa, and Kitchener-Waterloo commercial districts with dense buildings and negligible tree canopy.
- **Low-Low Clusters (Coldspots):** Algonquin Provincial Park, Georgian Bay forested shoreline, and Kawartha conservation zones acting as regional thermal sinks.
- **Spatial Outliers:** Rapidly developing suburban fringes (Low-High) and urban lakefront cooling corridors (High-Low).

8. Predictive Machine Learning with Spatial Cross-Validation

8.1 The Spatial Overfitting Problem

In conventional machine learning, standard random *k*-fold cross-validation randomly assigns points to training and test splits. When spatial autocorrelation is present, test observations are immediately adjacent to training observations, causing models to "memorize" geographic coordinates rather than learning functional relationships. This generates artificially inflated R^2 scores that collapse when applied to new geographic regions (Brenning, 2012; Schratz et al., 2021).

To solve this, we implement **Spatial Coordinate Block Cross-Validation** using `mlr3spatiotempcv`, which partitions space into disjoint geographic blocks:

```
# Set up Spatiotemporal Task and Spatial Resampling in mlr3
library(mlr3)
library(mlr3spatiotempcv)
library(mlr3learners)

# Define spatiotemporal regression task
task_uhi <- as_task_regr_st(
  modeling_data,
  target = "HIE_Index",
  coordinate_names = c("Longitude", "Latitude"),
  crs = "EPSG:4326"
)

# Coordinate-based spatial 5-fold cross-validation
resampling_spatial <- rsmp("spcv_coords", folds = 5)
```

8.2 Model Benchmark & Comparative Results

We benchmarked three diverse algorithm families across both standard random cross-validation and spatial coordinate cross-validation:

Model Architecture	Standard CV R^2 (Biased)	Spatial CV R^2 (True Generalization)	Spatial RMSE (°C)	Spatial MAE (°C)
Spatial Random Forest (ranger, 500 trees)	0.892	**0.764**	**0.542**	**0.418**
Regularized Elastic Net (glmnet, α = 0.5)	0.741	0.683	0.628	0.495
Weighted *k*-Nearest Neighbors (kknn, *k* = 7)	0.825	0.612	0.715	0.564

Finding: Standard random cross-validation overestimated predictive accuracy by 12.8% in Random Forest and by 21.3% in *k*-NN. The Spatial Random Forest achieved the highest true generalization performance (Spatial R^2 = 0.764, RMSE = 0.542 °C).

8.3 Permutation Feature Importance

Permutation-based mean squared error (MSE) loss extraction highlights the dominant structural drivers of Ontario's Urban Heat Island effect:

Rank	Predictor Variable	Relative Importance (%)	Microclimatic Impact Mechanism
1	**Mean NDVI (Zonal 2.5km)**	38.4%	Evapotranspirative vegetative cooling; canopy shading
2	**Building Footprint Density**	26.7%	Thermal mass storage in concrete; radiative heat trapping
3	**Percent Urban Impervious Cover**	18.9%	Low surface albedo; rapid solar absorption; absence of soil moisture
4	**Census Population Density**	9.2%	Anthropogenic emissions from transportation, HVAC, and industry
5	**Latitude / Geographic Position**	6.8%	Macro-scale solar zenith angle and regional climate background

9. Actionable Urban Planning & Policy Implications

The empirical findings translate into direct municipal climate adaptation policies for Ontario urban centers (Toronto, Mississauga, Brampton, Hamilton, Ottawa):

- **Tree Canopy Threshold Mandate:** Partial dependence profiling reveals that municipal wards maintaining ≥ 38% vegetative canopy cover (NDVI ≥ 0.45) experience an average 2.1 °C reduction in localized thermal excess.
- **High-Albedo Cool Roof Bylaws:** Commercial logistics warehouses and multi-residential complexes should mandate high solar reflectance surfaces (SRI ≥ 78) to curb thermal absorption.
- **Targeted Cooling Infrastructure:** LISA High-High hotspot clusters delineate priority zones for municipal hydration hubs, shade structures, and climate-resilient community cooling centers.

10. References

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